Joint Optimal and Adaptive 2-D Spatial Filtering Technique for FDA-MIMO SAR Deception Jamming Separation and Suppression

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***Abstract*— The existence of mainlobe deception jamming may seriously affect the information reconnaissance of synthetic aperture radar (SAR), resulting in unpredictable misjudgment. The similarity between deception jamming and real target scene in multiple domains brings excellent challenges to jamming detection and suppression. We find that the additional degrees of freedom (DOFs) brought by the multiple-input multiple-output (MIMO) frequency diversity array (FDA) system have unique advantages in solving this problem. Analysis proved that the coupling relationship of 2-D transmit–receive spatial frequency for the real target scene and the deception jamming is different. The 2-D transmit–receive spatial domain distribution of the real scene can be determined by the geometric configuration, while the distribution of deception jamming is inconsistent with the geometric position of the false targets due to the additional modulation. We proposed a joint optimal and adaptive 2-D spatial filtering technique based on this characteristic. The range-dependent phase of the echo is first compensated to separate the deception jamming from the real scene in the 2-D transmit–receive spatial domain. Then, jamming detection and localization are performed on the 2-D transmit–receive spatial spectrum. Adaptive covariance matrix estimation of the real target scene is performed from the range cells with no jamming detected. Afterward, the 2-D spatial filter is designed by solving a constrained optimization problem (COP) using the differential evolution (DE) algorithm based on the constrained dominance principle (CDP). Finally, jamming-free SAR imagery is obtained after processing all the pulses. The proposed technique exhibits sufficient performance in deception jamming rejection for SAR images, while the real target scene is almost undistorted. Simulation results are presented to demonstrate the performance of the proposed technique.**

***Index Terms*— Adaptive filtering, deception jamming, frequency diversity array (FDA), multiple-input multiple-output (MIMO), optimization problem, synthetic aperture radar (SAR).**

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I. Introduction

AS AN essential means of remote sensing, synthetic aperture radar (SAR) is widely used in geoscience information acquisition, such as vegetation classification, soil moisture observation, etc., owing to its ability of high-resolution imaging regardless of weather and sunlight [1].

Conventional SAR imaging and processing techniques usually assume that the electromagnetic environment of the observation scene is clean [2]. However, recent developments in the field of radar electronic countermeasure (ECM) technology have posed a considerable challenge to the sensing of SAR [3], [4]. Consequently, jamming rejection is progressively valued in SAR applications. Unlike radio frequency interference (RFI) and barrage jamming, deception jamming makes full use of the coherence of radar signals, hindering the information acquisition of radar with a much lower power cost [5]. Therefore, the interference suppression method based on power difference and signal independence is not applicable [6], [7]. By modulating and retransmitting the intercepted SAR signal, the jammer can introduce high fidelity and focused false targets into the SAR image to confuse the "vision" of the radar [8], [9]. Deception jamming exhibits similar features with the real scene echoes in multiple domains, which is challenging to identify and suppress accurately. Therefore, the deception jamming rejection in SAR imaging is still a knotty problem.

The essence of jamming countermeasures is to suppress the jamming signal in a separable domain by utilizing or constructing the difference between the real and jamming targets. This task is gradually difficult to complete in the time [10], frequency [11], and other common transform domains [12] with the development of jamming algorithm and the improvement of the real-time processing capability of digital jammers [13]. The premise of some traditional deception jamming countermeasures may not be satisfied, for example, the agile waveform requires that the jamming signal lags at least one pulse behind the real target [14].

As an important research direction to cope with deception jamming, the multichannel radar system improves the antijamming capability by increasing the spatial additional

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degrees of freedom (DOFs). Unlike aerial detection radar, the scattering characteristics of SAR application scenarios are usually complex, especially in land sensing, where the spatial location of the jamming overlaps highly with the real scene. Therefore, when the spatial notch method based on adaptive beamforming is applied to SAR processing, it will suppress the real target scene in the notch area at the same time, causing a significant loss of SAR image information [15]. Jamming suppression methods based on the blind source separation technique have a system DOF requirement of at least twice the number of signal sources, which is also difficult to meet for SAR imaging applications where the observation area is full of scatterers [16]. The spatial phase information processing of multichannel echoes is an effective method for multichannel SAR to suppress deception jamming, such as dual-channel cancellation [17] and space–time adaptive processing (STAP) [18]. However, the separable capability brought by the DOFs of the phased array still has boundaries. For instance, STAP is only applicable when false targets have additional Doppler modulation while losing efficiency for jamming exclusively modulated by time delay.

Antonik first proposed the frequency diversity array (FDA) in 2006 [19]. Since then, researchers have carried out extensive research on the scanning characteristics [20], data processing methods [21], beam pattern characteristics, and design methods [22] of FDA. Novelly, FDA introduces additional frequency increments between adjacent channels, leading to a range-angle coupled beam pattern. Benefit from this characteristic, FDA has been widely used in range and angle positioning [23], radio frequency stealth [24], and range ambiguity clutter suppression [25]. However, there are still many key issues worth considering in the application of actual radar systems because of the time-varying characteristic of the coherent FDA.

Sammartino et al. [26] proposed to combine FDA with multiple-input multiple-output (MIMO) radar. Later, the potential of such combined system in jamming suppression has sparked extensive thinking. In [27], a data-independent beamforming method is proposed to suppress false targets, which lag several pulses behind the true target. In [28], subspace projection technology is used to simultaneously suppress the clutter and jamming. In [29], the STAP method is extended to three dimensions by introducing the transmit spatial frequency, which realizes the effective detection of moving targets under the condition of dense false target jamming. The problem tackled by the above methods is the degradation of the traditional STAP performance caused by the disturbance of deception jamming. The simulated annealing algorithm is applied to the optimization of the nonlinear frequency offsets between the array elements in [30], and a single-peak beam pattern is obtained for jamming suppression. However, this method can only deal with the range deception jamming with a delay of more than one pulse repetition interval (PRI).

The purpose of the above STAP-based methods is to detect a single moving target, which is not applicable in SAR stationary scene imaging. In STAP methods [31], the adaptive filter is used to suppress clutter caused by static scenes, resulting in a notch at the clutter ridge. The clutter output power is represented by the estimated covariance matrix, and the moving target output is guaranteed by the steering vector constraints. However, for the problem faced in this article, stationary scenes considered as "clutter" are the targets to be preserved [32]. Since the spatiotemporal distribution of the deception jamming is highly coincident with the real scene echoes, it is difficult to obtain pure jamming echoes to estimate the covariance matrix. Consequently, the filter model of conventional STAP methods cannot be directly applied to jamming suppression in SAR imaging. The special problem of the FDA-MIMO SAR system in deception jamming suppression needs to be solved.

Our work considers the issue of deception jamming suppression for SAR imaging. The generated jamming targets are usually not in the exact location as the deception jammer to produce a confusing effect. Therefore, the spatial DOFs brought by FDA-MIMO can be used to separate the false targets and real target scene by filtering in the 2-D transmit–receive spatial frequency domain. In view of this separability, a joint optimal and adaptive 2-D spatial filtering technique for FDA-MIMO SAR deception jamming separation and suppression is suggested. The key innovations of our work are listed below.

1) A complete processing flow including preprocessing, deception jamming detection, localization, and suppression based on FDA-MIMO SAR is proposed. The separation mechanism and method of deception jamming and real target scene are given. Based on the proposed procedure, the deception jamming in the SAR image is effectively filtered out without using prior knowledge.

2) A joint optimal and adaptive 2-D spatial filtering technique suitable for SAR imaging is proposed. Based on the adaptively estimated covariance matrix of the real scene echoes and the detected jamming steering vectors, the designed 2-D filter exhibits considerable performance in deception jamming suppression with real targets almost undistorted.

The content of this article includes the following. Section II begins by presenting the signal model of FDA-MIMO SAR and elaborates on the deception jamming model. The theory and methodology of the proposed technique are introduced in Section III. Section IV gives the simulation to testify to the validity of the prospective technique, and the conclusions are obtained in Section V.

II. Signal Model of FDA-MIMO SAR

The FDA-MIMO SAR echo model of the real scene is first given in this section. Then, we clarify the assumptions we adopt for deception jamming, and the signal model is derived for jamming echo.

A. Signal Model of FDA-MIMO SAR

This article considers a collocated FDA-MIMO SAR. The array element combination configuration is set to N transmit channels and M receive channels. Select the first element along the platform course as a reference. According to the concept of FDA [19], carrier frequency increment is introduced

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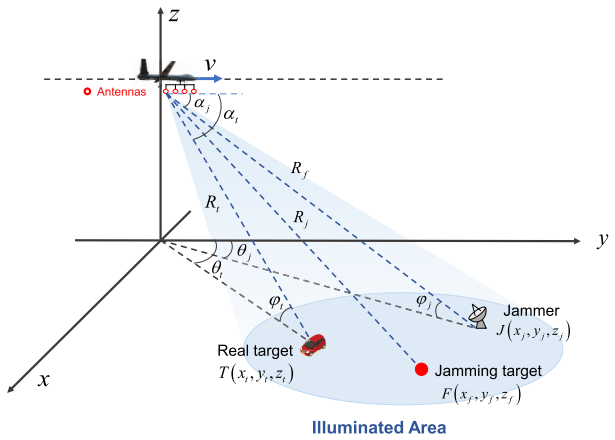


Fig. 1. Geometric configuration of the FDA-MIMO SAR system.

between adjacent elements. Then, the frequency transmitted by the nth element is given by

$$f_n = f_0 + (n-1)\Delta f, \quad n = 1, \ldots, N \tag{1}$$

where Δf is the frequency offset between elements, f_0 is the reference carrier frequency of the SAR transmit signal, and $\Delta f \ll f_0$. Then, the transmit signal of the nth element is

$$V_n(t) = \text{rect}\big(t/T_p\big)U_n(t)\exp(j2\pi f_n t), \quad n = 0, 1, \ldots, N-1 \tag{2}$$

where rect$(\cdot)$ is the rectangular function, $U_n(t)$ represents the transmitted waveform, and T_p is the pulse duration. To separate the echoes of different transmit elements at the receiving end, we assume that $U_n(t)$ is a group of waveforms with ideal orthogonal performance [29], namely

$$\int_T U_{n_1}(t) \cdot U_{n_2}^*(t-\tau)dt = 0, \quad \text{when} \quad n_1 \neq n_2 \tag{3}$$

where $*$ and τ denote the conjugate operator and time delay, respectively.

Fig. 1 gives the geometrical configuration of the FDA-MIMO SAR system. Assuming that the platform moves parallel to the y-axis at speed v. We take the side-looking SAR as an example to analyze in this article. The jammer is located at point $J(x_j, y_j, z_j)$ within the mainlobe illumination area. Point $F(x_f, y_f, z_f)$ represents the preset position where the jammer generates the jamming targets. Without loss of generality, take a target at point $T(x_t, y_t, z_t)$ as an example to establish the echo model. Backscattered by the far-field target, the transmitted signals of N elements are received by M elements. After down-conversion, the real target echo of the mth received element is

$$\begin{aligned} S_m(t,\eta) = &\sum_{n=1}^{N} \sigma_i \text{rect}\left(\frac{t-\tau_{nm}(\eta)}{T_p}\right) U_n(t-\tau_{nm}(\eta)) \\ &\times \exp[-j2\pi f_0 \tau_{nm}(\eta)] \\ &\times \exp[j2\pi(n-1)\Delta f(t-\tau_{nm}(\eta))] \end{aligned} \tag{4}$$

where t is the fast time, η is the slow time, and σ_i is the scattering coefficient of the real target. $\tau_{nm}(\eta)$ is time delay of the echo, which is given by

$$\begin{aligned} \tau_{nm}(\eta) &= \frac{2R_{nm}(\eta)}{c} \\ &= \frac{2R_t(\eta)}{c} - \frac{d(n-1)\cos(\alpha_t(\eta))}{c} \\ &\quad - \frac{d(m-1)\cos(\alpha_t(\eta))}{c} \end{aligned} \tag{5}$$

where c is the light speed and d is the element spacing. $R_{nm}(\eta)$ is the range history of the echo from the nth transmit element to the mth receive element. $R_t(\eta)$ and $\alpha_t(\eta)$ are the slant range and conical angle between the target and the reference element, respectively. The cosine of the angle $\alpha_t(\eta)$ is given by

$$\cos(\alpha_t(\eta)) = \cos(\theta_t(\eta))\cos(\varphi_t(\eta)) \tag{6}$$

where $\theta_t(\eta)$ and $\varphi_t(\eta)$ denote the azimuth angle (included angle with the forward direction of the platform) and elevation angle of the target T. Benefiting from the DOFs introduced by MIMO, for orthogonal waveforms, the echoes corresponding to different transmit channels can be separated according to different reference functions when matched filtering. The matched filtering output corresponding to the combination of nth and mth transceiver elements is given by

$$\begin{aligned} S_{nm}(t,\eta) = &\sigma_i w_{r,n}(t-\tau_{nm}(\eta))w_a(\eta) \\ &\times \exp[-j2\pi f_0 \tau_r(\eta)] \\ &\times \exp[(m-1)d\cos(\alpha_t(\eta))/\lambda] \\ &\times \exp[-j2\pi(n-1)\Delta f \tau_r(\eta)] \\ &\times \exp\left[j2\pi \frac{\Delta f(n-1)^2 d\cos(\alpha_t(\eta))}{c}\right] \\ &\times \exp\left[j2\pi \frac{\Delta f d(n-1)(m-1)\cos(\alpha_t(\eta))}{c}\right] \end{aligned} \tag{7}$$

where $w_{r,n}(t-\tau_{nm}(\eta))$ is the range envelope of the nth matched filter output, and $w_a(\eta)$ denotes the envelope in azimuth. $\tau_r(\eta) = 2R_t(\eta)/c$ is the time delay of the reference channel. To further simplify, $w_{r,n}(t-\tau_{nm}(\eta))$ is approximated as $w_{r,n}(t-\tau_r(\eta))$ since $R_{nm}(\eta) \approx R_t(\eta)$. The last two quadratic terms are negligible when $(N-1)\Delta f \ll f_0$ [26]. Therefore, (7) is matched as

$$\begin{aligned} S_{nm}(t,\eta) \approx &\sigma_i w_{r,n}(t-\tau_r(\eta))w_a(\eta) \\ &\times \exp[-j2\pi f_0 \tau_r(\eta)] \\ &\times \exp[(m-1)d\cos(\alpha_t(\eta))/\lambda] \\ &\times \exp[j2\pi(n-1)(-\Delta f \tau_r(\eta) + d\cos(\alpha_t(\eta))/\lambda)]. \end{aligned} \tag{8}$$

To analyze and process the spatial information of each pulse, the range compressed echoes of each element are stacked into an $NM \times 1$ snapshot data matrix. Then, the echo of the range cell, which contains the target, is given by

$$\begin{aligned} \mathbf{S}_t &= [s_{11}, s_{21}, \ldots, s_{N1}, s_{12}, \ldots, s_{nm}, \ldots, s_{NM}]^T \\ &= \varepsilon_{\mathbf{s}} \mathbf{B}_r(\alpha_t(\eta)) \otimes \mathbf{B}_t(R_t(\eta), \alpha_t(\eta)) \end{aligned} \tag{9}$$

where ε_s is the complex amplitude, and $\otimes$ is the Kronecker product. $\mathbf{B}_t(R, \alpha) \in \mathbb{C}^{N\times 1}$ is the steering vector of the transmit

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frequency dimension and $\mathbf{B}_r(\alpha) \in \mathbb{C}^{M\times 1}$ is the steering vector of the receive frequency dimension, which is defined as

$$\mathbf{B}_t(R,\alpha) = \left[1, e^{j2\pi f_t}, \ldots, e^{j2\pi(N-1)f_t}\right]^T \tag{10}$$

$$\mathbf{B}_r(\alpha) = \left[1, e^{j2\pi f_r}, \ldots, e^{j2\pi(M-1)f_r}\right]^T \tag{11}$$

where $f_t = d\cos\alpha/\lambda - 2R\Delta f/c$ and $f_r = d\cos\alpha/\lambda$ are defined as the transmit and receive spatial frequency, respectively. f_t is range-angle coupled due to the small frequency increments at the transmitter. Consequently, radar can establish a relationship with any point in the observation scene through a positioning form similar to polar coordinates. The spatial information brought by the increase of DOFs provides a theoretical basis for the subsequent filtering of deception jamming in the 2-D $f_t - f_r$ domain.

B. Signal Model of Deception Jamming

To facilitate our discussions, the radar parameter estimation error, quantization error, and other factors faced during jamming generation are not considered, which means the deception jamming accurately meets the SAR imaging focusing requirement. The additional time delay introduced by the jammer is not limited, i.e., the modulation and retransmission within one PRI are allowed. These assumptions guarantee the effectiveness of the proposed model even as the deception jamming technology advances.

The jammer is assumed to be located at point J. The radar signal received at the jammer is given by

$$\begin{aligned} V_j(t,\eta) &= \sum_{n=1}^{N} V_{jn}(t,\eta) \\ &= \sum_{n=1}^{N} \mathrm{rect}\left(\frac{t-\tau_{jn}(\eta)}{T_p}\right) U_n\left(t-\tau_{jn}(\eta)\right) \\ &\quad \times \exp\left[j2\pi f_n\left(t-\tau_{jn}(\eta)\right)\right] \end{aligned} \tag{12}$$

where $\tau_{jn}(\eta) = R_j(\eta)/c - (n-1)d\cos(\alpha_j(\eta))/\lambda$ represents the time delay from the nth element to the jammer. $R_j(\eta)$ represents the slant range between the jammer and reference element. $\alpha_j(\eta)$ represents the conical angle of the jammer. The cosine of the conical angle can also be obtained from the azimuth angle θ_j and elevation angle φ_j by calculating $\cos(\alpha_j(\eta)) = \cos(\theta_j(\eta))\cos(\varphi_j(\eta))$. According to the acquired radar position and motion parameters, the deception jammer modulates and forwards the SAR signal to form the jamming target at the preset position F. The generation of the jamming target can be described as

$$\begin{aligned} V_j'(t,\eta) = \sigma_f V_j(t,\eta) * \delta\left(t-\Delta T_j(\eta)\right) \\ \times \exp\left[-j2\pi f_n \Delta T_j(\eta)\right] \end{aligned} \tag{13}$$

where σ_f is the amplitude modulation factor related to the scattering coefficient, $\delta(\cdot)$ is the unit impulse function, and $\Delta T_j(\eta) = 2[R_f(\eta) - R_j(\eta)]/c$ is the time delay modulated by the jammer. The modulated jamming signal is retransmitted to the radar receiver, also received by M elements. After down-conversion and matched filtering, the jamming echo of the combination of nth and mth transceiver elements is

$$\begin{aligned} &S_{jnm}(t,\eta) \\ &\quad \approx \sigma_f w_{r,n}\left(t-\tau_f(\eta)\right) w_a\left(\eta-\eta_j\right) \\ &\qquad \times \exp\left[-j2\pi f_0 \tau_f(\eta)\right] \\ &\qquad \times \exp\left[(m-1)d\cos\left(\alpha_j(\eta)\right)/\lambda\right] \\ &\qquad \times \exp\left[j2\pi(n-1)\left(-\Delta f\,\tau_{rj} + d\cos\left(\alpha_j(\eta)\right)/\lambda\right)\right] \end{aligned} \tag{14}$$

where η_j is the time delay of the jammer in the azimuth domain, $\tau_f(\eta) = 2R_f(\eta)/c$ and $\tau_{rj}(\eta) = 2R_j(\eta)/c$ are the time delay from the reference channel to the jamming target F and the jammer, respectively. After stacking the data of different elements, the echo of the range cell containing the jamming target is given by

$$\begin{aligned} \mathbf{S}_j &= \left[s_{j11}, s_{j21}, \ldots, s_{jN1}, s_{j12}, \ldots, s_{jnm}, \ldots, s_{jNM}\right]^T \\ &= \varepsilon_f \mathbf{B}_r\left(\alpha_j(\eta)\right) \otimes \mathbf{B}_t\left(R_j(\eta), \alpha_j(\eta)\right). \end{aligned} \tag{15}$$

Compared with the echo model in Section II-A, the jammer simulates the same range history as the real target at F through time and Doppler modulation so that the jamming echo can be focused and appear at the preset position in the SAR image. However, the phase difference between elements is caused by the spatial position of the signal transmission path. Thus, the second and third phase terms in (14) contain the spatial information of the jammer, i.e., R_j and α_j, which cannot be falsified by jamming modulation. Utilizing such a feature, we can separate and suppress the deception jamming in the spatial domain.

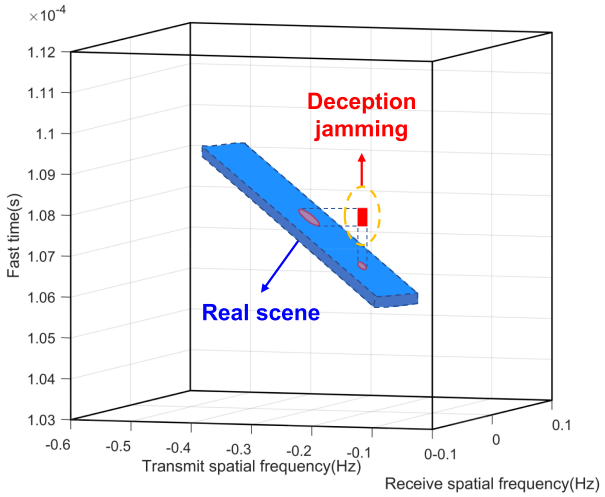


Fig. 2. 3-D space–time coupling relationship of real target scene and deception jamming.

Fig. 2 shows the 3-D space–time coupling relationship of a simulated scene and deception jamming. The feature difference analyzed above is visualized in the 3-D space–time joint domain. The blue and red area represents the space–time distribution of the real target scene and deception jamming, respectively. The projection of the deception jamming in the time or space domain is aliased with the real scene. Therefore, it is extremely difficult to separate the two using a single feature domain. This high-dimensional separable feature is the basis of the deception jamming separation and suppression technique proposed in this article.

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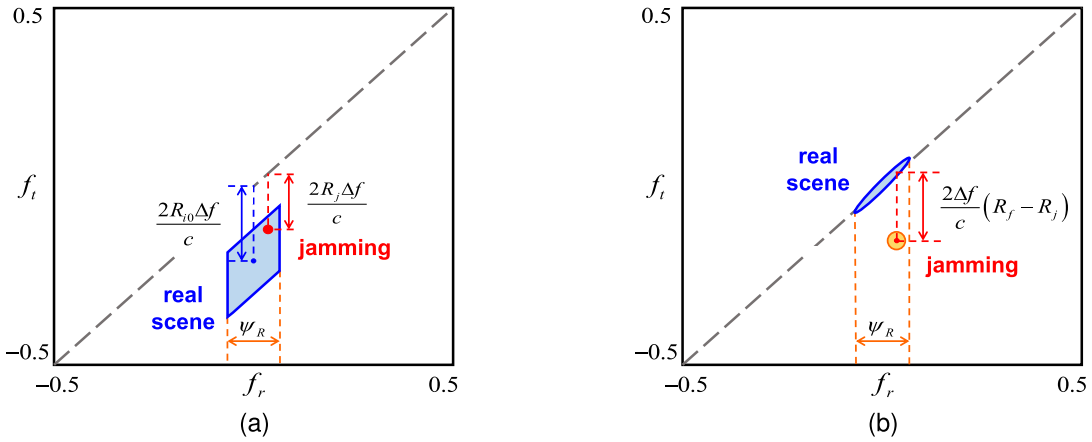


Fig. 3. Sketch of 2-D transmit–receive spatial frequency domain distribution of real scene and deception jamming. (a) Before phase compensation. (b) After phase compensation.

III. Principle of the Proposed Technique

The coupling relationship of 2-D transmit–receive spatial frequency for the deception jamming and the real target scene is first analyzed. Then, the range-dependent phase compensation is performed to separate the deception jamming from the real scene in the 2-D spatial domain. Furthermore, the jamming detection and localization of the 2-D transmit–receive spatial spectrogram are carried out using morphological operations. The methodology for the joint optimal and adaptive 2-D spatial filter design is presented in detail. Finally, the complete processing flow of deception jamming suppression in SAR imaging is given.

A. Target-Jamming Separation

The additional spatial DOFs introduced by multichannel frequency diversity increase the separable dimension of deception jamming and the real target scene. To achieve the target-jamming separation, the 2-D spatial frequency coupling relationship of the echo is analyzed. The transmit and receive spatial frequencies are defined as

$$f_t(R,\alpha) = \frac{d\cos\alpha}{\lambda} - \frac{2R\Delta f}{c} \tag{16}$$

$$f_r(\alpha) = \frac{d\cos\alpha}{\lambda} \tag{17}$$

where R and α are the slant range and conical angle of the echo incident position. An illustration of the real scene and deception jamming in the 2-D spatial frequency domain is shown in Fig. 3(a). The real scene echo (blue area) comes from the scatterers at different positions within the SAR beam illumination area according to the geometric configuration. The receive spatial frequency bandwidth can be expressed as

$$\psi_R = d\sin(\theta_{bw})/\lambda \tag{18}$$

where θ_{bw} is the radar azimuth beamwidth. If the squint angle is not 0, there will be an additional frequency shift in the direction parallel to the diagonal. The echoes of different range cells are shifted relative to the diagonal due to the range-dependent term in (16). R_{i0} is the slant range of the central point of the illuminated area. The transmit spatial frequency bandwidth is determined by the strip width of SAR. The red dot indicates the jammer's position and R_j represents the slant range of the jammer. The spatial frequency domain distribution of the mainlobe deception jamming coincides with the real scene, which cannot be directly separated.

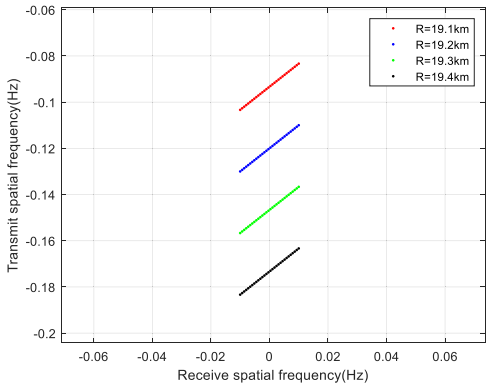


Fig. 4. 2-D spatial spectrum of echoes respect to different range cells.

Since the beam direction of strip map SAR remains unchanged, the echoes of different pulses are distributed in the same position in the 2-D spatial domain. Fig. 4 gives the 2-D spatial frequency coupling relationship of echoes with respect to different range cells. The simulation parameters are given as follows: $(x_r, y_r, z_r) = (7000, 0, 5000)$ m, $PRF = 1000$ Hz, $V = 200$ m/s, and $\lambda = 0.03$ m. The echo of one range cell is distributed along the direction parallel to the diagonal, and the offset relative to the diagonal increases with the range.

Using Capon spectral estimation method [33], the minimum variance estimation of the 2-D transmit–receive spatial frequency spectrum of the echoes can be obtained by traversing the steering vector corresponding to each grid point in the plane. The power spectrum can be expressed as

$$\mathbf{P}_{MVM}(f_r, f_t) = \frac{1}{\mathbf{B}_{tr}^H(R,\alpha)\mathbf{R}^{-1}\mathbf{B}_{tr}(R,\alpha)} \tag{19}$$

where $\mathbf{R}^{-1}$ is the inverse of the echo covariance matrix, and H is the conjugate transpose operator. $\mathbf{B}_{tr}(R,\alpha)$ represents the 2-D spatial frequency steering vector, which is defined as

$$\mathbf{B}_{tr}(R,\alpha) = \mathbf{B}_r(\alpha) \otimes \mathbf{B}_t(R,\alpha). \tag{20}$$

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To separate the real targets from the deception jamming, the energy of the real target scene in the 2-D spatial domain is expected to be concentrated as much as possible. For SAR imaging, the range-dependent phase term in (16) can be compensated according to the SAR geometric configuration. Since the range-dependent phase is associated with the transmit frequency offset, the compensation phase of the nth transmit element's echo at the kth range cell can be described as

$$p_{cnk} = \exp\left[j2\pi(n-1)\frac{2R_k}{c}\right] \tag{21}$$

where R_k is the slant range of the kth range cell. For an NM-dimensional snapshot of a pulse, the compensated phase vector of the kth range cell is given by

$$\mathbf{C}(R_k) = \left[1, e^{j4\pi\Delta f R_k/c}, \ldots, e^{j4\pi\Delta f(N-1)R_k/c}\right]^T. \tag{22}$$

Then the compensated snapshot vector can be expressed as

$$\hat{\mathbf{S}}_k = \mathbf{S}_k \circ \mathbf{C}(R_k) \tag{23}$$

where $\mathbf{S}_k = [s_{11}^k, s_{21}^k, \ldots, s_{NM}^k]^T$ is the echo vector of the kth range cell of a pulse and $\circ$ is the Hadamard product.

The jammer changes the range and Doppler information of the intercepted signal from the jammer position (x_j, y_j, z_j) to the false target position (x_f, y_f, z_f) through modulation, while the spatial information of jamming echoes still reflects the jammer's position. The deception jamming signal is compensated by the phase corresponding to the false target position when compensating the range-dependent phase based on geometric configuration. The mismatch between the time and spatial information of the deception jamming results in a residual phase after range-dependent compensation. Thus, the transmit spatial frequencies of the real scene and deception jamming after compensation turn to

$$\hat{f}_{ts}(\alpha_t) = \frac{d\cos\alpha_t}{\lambda} \tag{24}$$

$$\hat{f}_{tj}\left(R_j - R_f, \alpha_j\right) = \frac{d\cos\alpha_j}{\lambda} - \frac{2\Delta f}{c}\left(R_j - R_f\right). \tag{25}$$

Fig. 3(b) shows the sketch of the 2-D transmit–receive spatial spectrum after phase compensation. The real scene echoes in different range cells are compensated to the diagonal, while the deception jamming is still outside the diagonal due to the residual phase. As a result, the jamming and real scene echoes are separated in the 2-D spatial domain. It is worth mentioning that due to the different compensation phases of different range cells, if the false target spans more than one range cell, the jamming will extend along the transmit frequency dimension after compensation.

B. Jamming Detection and Localization

Without prior knowledge, jamming detection and feature extraction are essential prerequisites for jamming suppression. Although the features in multiple domains coincide with the real target scene, the deception jamming is separable in the 2-D spatial frequency domain after range-dependent phase compensation. Therefore, a method for deception jamming detection of each range cell is proposed based on the morphological operation of the 2-D spatial spectrogram. According to the detection result, the 2-D spatial steering vectors of the jamming can be further obtained. Then, by extracting the echo data of the range cells without jamming, the covariance matrix of the real target scene is estimated. The specific procedures of jamming detection and localization are given as follows.

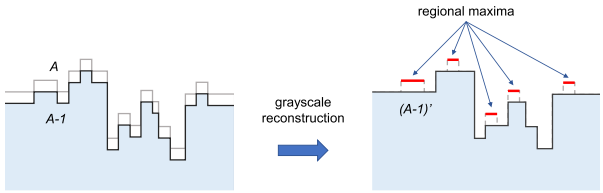


Fig. 5. Schematic of extracting regional maxima by morphological grayscale reconstruction method.

To eliminate the influence of noise on the detection results, an energy threshold is set to discard the noise-only range cells. The sum of the energy mean and standard deviation of all range cells is taken as the upper energy threshold for selecting noise cells [34], which is given as

$$Th = E_m + \left[\sum_{k=1}^{K}\frac{(E_k - E_m)^2}{K-1}\right]^{\frac{1}{2}} \tag{26}$$

where E_k is the energy of the kth range cell (assuming there is a total of K range cells) and E_m is the average energy of all range cells. Range cells below this threshold are determined as noise-only range cells. With the noise-only range cells, the power level of the noise can be estimated further to eliminate the effect of noise in jamming detection. In the subsequent processing, only range cells with energy above the threshold will be considered.

For range cells containing echoes, the data of a certain number of range cells can be extracted by sliding-window sampling. According to the Reed-Mallett-Brennan (RMB) rule [35], the sample numbers in a sliding window are required to be at least twice the channel DOFs of the system, i.e., the window width $l_w \geq 2NM$. Then, the 2-D spatial spectrum $\mathbf{P}_{MVM}$ of the extracted range cells is estimated by the Capon method. Based on the distribution characteristics of the deception jamming analyzed in Section III-A, the key to jamming detection is to detect whether there are peaks outside the diagonal. Morphological grayscale reconstruction provides an efficient way to extract regional maxima of grayscale images [36]. This method does not rely on prior knowledge and needs no threshold. Even if the jamming spectral peaks are pretty weak relative to the real scene echo, they can be extracted as regional maxima. As shown in Fig. 5, given an original image A, first perform grayscale morphological reconstruction on its offset $A-1$ shifted down by 1 grid to obtain $(A-1)'$. Then, the regional maxima of A can be obtained by deducting $(A-1)'$ from A. The binary image $M(A)$ of the regional maxima of A is given by

$$M(A) = A - \rho_A(A-1). \tag{27}$$

The reconstruction $\rho_A(A-1)$ of mask A from marker $A-1$ can be expressed as

$$\rho_A(A-1) = \bigcup_{A-1\cap A_k\neq 0} A_k \tag{28}$$

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where $A_1, A_2, \ldots, A_n$ are the connected components of A. After grayscale morphological reconstruction, the binary image of the detection results can be obtained, in which the output of the regional maximum pixel of the image is 1, and the rest is 0. Based on the previous analysis, there are only two cases for the regional maximum distribution of the 2-D spatial spectrum: the real target scene on the diagonal and the deception jamming outside the diagonal. Therefore, the jamming can be detected by judging whether regional maxima are outside the diagonal line. Assuming that a total of I regional maxima are detected, which can be described as

$$Q = \{q_i | M(q_i) = 1\}, \quad i = 1, 2, \ldots, I \tag{29}$$

where q_i represents the ith detected regional maximum. The deception jamming detection problem can be transformed into the following binary decision problem:

$$\begin{cases} \text{Jamming}, & q_i(f_{ri}, f_{ti}) \in Q,\ f_{ri} \neq f_{ti} \\ \text{Real target}, & q_i(f_{ri}, f_{ti}) \in Q,\ f_{ri} = f_{ti}. \end{cases} \tag{30}$$

Collect all the off-diagonal regional maxima and derive their coordinates. Then, the steering vector $\hat{\mathbf{s}}_{tr\,j_i}(\Delta r_i, \alpha_i)$ can be calculated from the coordinates (f_{ri}, f_{ti}). With the spectral location of the deception jamming, we can filter the jamming signal in a targeted manner. After traversing all range cells according to the step size, the range cells with deception jamming detected are eliminated. The real scene echoes in the remaining range cells satisfy the independent and identically distributed (I.I.D) condition [35], which can be used as the sample source for covariance matrix estimation of the real target scene. As a result, the deception jamming is detected and localized in the 2-D spatial domain without prior knowledge. Besides, the jamming and real scene information required for the 2-D filter design problem has been adaptively obtained.

C. Joint Optimal and Adaptive 2-D Filter Design

After range-dependent phase compensation, the deception jamming and real scene are effectively separated in the 2-D spatial domain. Therefore, designing a suitable 2-D filter can effectively reject jamming with the real scene almost undistorted. According to the characteristics of SAR echo and deception jamming, we propose a joint optimal and adaptive model to solve the filter response. The output power of the real target scene is adaptively determined by the covariance matrix estimated by the real scene echo, while the notch at the deception jamming is optimized by setting constraints according to the detection results.

The sketch of the ideal 2-D spatial filter response is shown in Fig. 6. The upper and lower parts describe the corresponding relationship between the ideal filter response and the echo spectrum. To suppress deceptive jamming and protect the real targets, the filter is expected to maintain a distortion-free gain of 1 at the position of the real echo and form a notch as deep as possible at the position of the deception jamming.

For the NM dimension echo of a pulse, the filter weight $\mathbf{w}$ to be designed can be expressed as

$$\mathbf{w} = [w_1, w_2, \ldots, w_{NM}]^T. \tag{31}$$

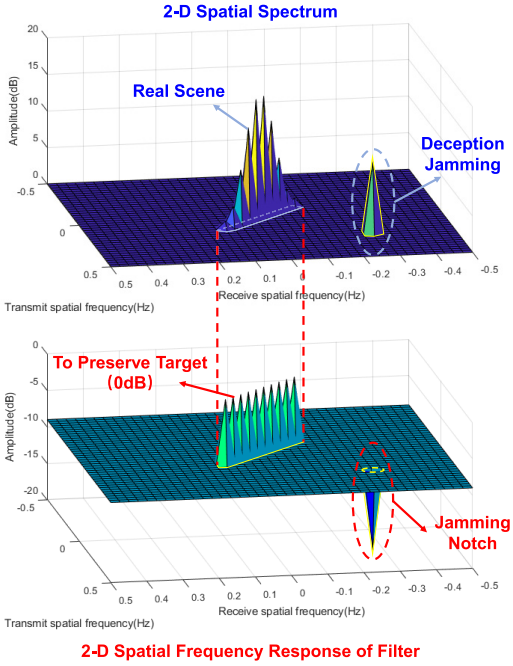


Fig. 6. Sketch of the 2-D spatial filter design.

The 2-D spatial frequency filter response is given by

$$L(f_r, f_t) = \mathbf{w}^H \mathbf{B}_{tr}(f_r, f_t). \tag{32}$$

By designing the weight $\mathbf{w}$, the output gain of each point, i.e., the filter shape, can be adjusted. The filter output of the multichannel echo of one pulse is given by

$$G(t) = \mathbf{w}^H \hat{\mathbf{S}}(t) \tag{33}$$

where $\hat{\mathbf{S}}(t)$ is the echo after range-dependent phase compensation. Then, the average power of the output signal is calculated as

$$\begin{aligned} P &= E\left[G(t)G^H(t)\right] \\ &= \mathbf{w}^H E\left[\hat{\mathbf{S}}(t)\hat{\mathbf{S}}^H(t)\right]\mathbf{w} \\ &= \mathbf{w}^H \mathbf{R}_{\hat{S}} \mathbf{w} \end{aligned} \tag{34}$$

where $E[\cdot]$ represents the statistical average and $\mathbf{R}_{\hat{S}}$ is the autocovariance matrix of the echo after phase compensation. To effectively suppress deception jamming, the ratio of the average power of the real target scene to the jamming after filtering is expected to reach the maximum, which can be expressed as

$$\Gamma = \frac{P_s}{P_j} = \frac{\mathbf{w}^H \mathbf{R}_{\hat{S}_t} \mathbf{w}}{\mathbf{w}^H \mathbf{R}_{\hat{S}_j} \mathbf{w}} \tag{35}$$

where P_s is the average power of the real target scene and P_j is the average power of the deception jamming after filtering. $\mathbf{R}_{\hat{S}_t}$ and $\mathbf{R}_{\hat{S}_j}$ are the covariance matrices of those two, respectively. Using the jamming detection and data screening strategy proposed in Section III-B, $\mathbf{R}_{\hat{S}_t}$ is estimated from the echoes of the jamming-free range cells, which is given by

$$\mathbf{R}_{\hat{S}_t} = \frac{1}{N_k} \sum_{k=1}^{N_k} \mathbf{S}_{tk} \mathbf{S}_{tk}^H, k \in K_t \tag{36}$$

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where $\mathbf{S}_{tk}$ is the echo of the kth range cell of each pulse, K_t is the set of range cell indices for which no deception jamming is detected, and N_k is the total number of set elements. Thus, the filtered output average power of the real targets is adaptively determined by the real scene echo.

Since the jamming signal is aliased with the real target scene in the time domain, it is difficult to estimate $\mathbf{R}_{\hat{S}_j}$ directly from echoes. The spatial information of the jamming signal is composed of the 2-D steering vector and the complex amplitude, which is described as

$$\hat{\mathbf{S}}_j(t) = \hat{\mathbf{B}}_{trj}(f_{rd}, f_{td}) \circ \varepsilon_f(t) \tag{37}$$

where $\hat{\mathbf{B}}_{trj}(f_{rd}, f_{td})$ is the 2-D steering vector of the deception jamming after range-dependent phase compensation. f_{rd} and f_{td} are the coordinates of the detected jamming points. In the 2-D transmit–receive spatial frequency domain, $\hat{\mathbf{B}}_{trj}(f_{rd}, f_{td})$ determines the position of the deception jamming and ε_f only affects the amplitude of the spectrum. Therefore, $\mathbf{R}_{\hat{S}_j}$ in (35) is replaced by the covariance matrix of the steering vector to form a notch at the position of deception jamming, which is given by

$$\mathbf{R}_{B_j} = \frac{1}{N_j} \sum_{i=1}^{N_j} \hat{\mathbf{B}}_{trji}(f_{rdi}, f_{tdi}) \hat{\mathbf{B}}_{trji}^H(f_{rdi}, f_{tdi}) \tag{38}$$

where $\hat{\mathbf{B}}_{trji}(f_{rdi}, f_{tdi})$ is the 2-D steering vector of the ith jamming point detected and N_j is the number of jamming points. The output gain of the jamming notch is subject to an additional constraint to ensure minimum performance, and the ratio Γ is optimized for the best possible results. Then, using the optimization method, the filter response with desired performance can be solved. The 2-D spatial filter design problem is modeled as the following constrained optimization problem (COP):

$$\begin{aligned} \min \quad & F(\mathbf{w}) = -\frac{\mathbf{w}^H \mathbf{R}_{\hat{S}_t} \mathbf{w}}{\mathbf{w}^H \mathbf{R}_{B_j} \mathbf{w}} \\ s.t. \quad & \begin{cases} \left| \mathbf{w}^H \mathbf{S}_{ti}(f_{ri}, f_{ti}) - 1 \right| \le \xi, & f_{ri} = f_{ti} \\ \mathbf{w}^H \mathbf{S}_j \mathbf{w} \le \upsilon \end{cases} \end{aligned} \tag{39}$$

where ξ is an error tolerance and υ represents the upper limit of the jamming output gain. $\mathbf{S}_{ti}(f_{ri}, f_{ti})$ represents the steering vector of the real target scene area, where $f_{ri}, f_{ti} \in [-0.5d \sin(\theta_{bw})/\lambda, 0.5d \sin(\theta_{bw})/\lambda]$. The first constraint states that the filter response is approximately 1 at the real target scene area, which means that the real scene echo can be preserved after filtering. The second constraint limits the notch depth at the jamming position. The required filter weight $\mathbf{w}^*$ is the solution that minimizes the objective function $F(\mathbf{w})$ in (39).

For the above COP, we choose the differential evolution (DE) algorithm to solve it. The DE algorithm is a robust stochastic real-parameter optimization algorithm whose excellent computing performance has been proven in many international competitions [37]. Its comprehensive capability, including convergence speed, accuracy, and robustness, is competitive in many practical applications. Moreover, the implementation of DE is much simpler and more straightforward than most other evolutionary algorithms (EAs), which is suitable for engineering applications.

Algorithm 1 2-D Spatial Frequency Filter Optimization Algorithm

1: Obtain $\mathbf{R}_{\hat{S}_t}$ and $\mathbf{R}_{B_j}$ of the current pulse echo.
2: Set the constant error tolerance ξ and the upper limit υ, and construct the COP in (39).
3: Set the DE parameters: crossover rate Cr, population size N_P and scale factor F_s.
4: Set the boundaries of the parameter as $[c_{\min}, c_{\max}]$.
5: Initialize the generation number G and the population $\mathbf{P}_G = \{\mathbf{C}_{1,G}, \ldots, \mathbf{C}_{N_P,G}\}$, where $\mathbf{C}_{i,G} = [c_{1,i,G}, c_{2,i,G}, \ldots, c_{2NM,i,G}]^T$. Each individual is initialized uniformly distributed in the range $[c_{\min}, c_{\max}]$.
6: **While** the termination condition is not met
7: **Do**
8: **For** $i = 1$ to N_P
Generate the donor vector $\mathbf{V}_{i,G}$ of the ith target vector $\mathbf{C}_{i,G}$ according to the differential mutation scheme of DE
9: Generate the trial vector $\mathbf{U}_{i,G}$ of the ith target vector $\mathbf{C}_{i,G}$ through binomial crossover
10: Evaluate the trial vector $\mathbf{U}_{i,G}$
11: **If** $\mathbf{U}_{i,G}$ is feasible and $\mathbf{C}_{i,G}$ is infeasible, **then** $\mathbf{C}_{i+1,G} = \mathbf{U}_{i,G}$
12: **Elseif** $\mathbf{C}_{i,G}$ is feasible and $\mathbf{U}_{i,G}$ is infeasible, **then** $\mathbf{C}_{i+1,G} = \mathbf{C}_{i,G}$
13: **Elseif** $\mathbf{U}_{i,G}$ and $\mathbf{C}_{i,G}$ are both infeasible, but the constraint violation of $\mathbf{U}_{i,G}$ is smaller than $\mathbf{C}_{i,G}$, **then** $\mathbf{C}_{i+1,G} = \mathbf{U}_{i,G}$
14: **Elseif** $\mathbf{U}_{i,G}$ and $\mathbf{C}_{i,G}$ are both infeasible, but the constraint violation of $\mathbf{C}_{i,G}$ is smaller than $\mathbf{U}_{i,G}$, **then** $\mathbf{C}_{i+1,G} = \mathbf{C}_{i,G}$
15: **Elseif** $\mathbf{U}_{i,G}$ and $\mathbf{C}_{i,G}$ are both feasible and $F(\mathbf{C}_{i,G}) \le F(\mathbf{U}_{i,G})$, **then** $\mathbf{C}_{i+1,G} = \mathbf{C}_{i,G}$
16: **Else** $\mathbf{C}_{i+1,G} = \mathbf{U}_{i,G}$
17: **End if**
18: **End for**
19: $G = G + 1$
20: **End while**
21: Output the optimal solution $\mathbf{C}_G^*$ based on $F(\mathbf{C}_G^*) = \min F(\mathbf{C}_{i,G})$.

The constraint dominance principle (CDP) proposed by Deb [38] is applied with DE to handle the constraints. The specific criteria of CDP will not be repeated in this article, and interested readers can refer to [38]. By introducing CDP into the selection step of DE, the search can gradually lead to the optimum solution without the requirement of any penalty parameter. Since the filter weight $\mathbf{w}$ is an $NM \times 1$ complex vector, to adapt to the DE algorithm, the weight value is decomposed into real and imaginary parts, namely

$$w_i = a_i + jb_i, \quad i = 1, 2, \ldots, NM. \tag{40}$$

Then the $2NM \times 1$ real vector $\mathbf{C} = [a_1, a_2, \ldots, a_{NM}, b_1, \ldots, b_{NM}]^T$ is the parameter to be optimized. The specific steps of the proposed 2-D spatial filter optimization

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algorithm are given in the form of pseudocode. By implementing the CDP-DE algorithm, the 2-D matched filter weights for deception jamming suppression can be obtained.

D. FDA-MIMO SAR Deception Jamming Suppression

This section gives the processing flow of deception jamming suppression for SAR imaging with the proposed technique.

After matched filtering, the range-dependent phase of the echo received by multiple elements is compensated first. As the radar moves, the conical angle between the jammer and the radar will change slightly, causing spectrum extension of the jamming in the 2-D spatial frequency domain. To avoid the possible performance loss caused by spectrum extension, we divide the aperture time into sufficient periods as the filter processing interval (FPI). The pulses in an FPI are processed with a unified filter, whose weight is calculated by the pulse data $\hat{\mathbf{S}}(t, \eta_c)$ of the interval center η_c. Then, rearrange the echo data of multiple elements of each selected pulse for subsequent spatial processing. The morphologically based method is applied to detect the deception jamming for the range cells selected by the sliding window. The covariance matrices of the real target scene and the jamming steering vector are further extracted. Afterward, the 2-D spatial filter optimization design problem is solved by the CDP-DE algorithm. With the real solution $\mathbf{C}_G^*$, the required jamming suppression filter weight can be obtained from (40)

$$\mathbf{w}^* = \left[a_1^* + jb_1^*, a_2^* + jb_2^*, \ldots, a_{NM}^* + jb_{NM}^*\right]^T. \quad (41)$$

Then the echo data after filtering can be described as

$$\mathbf{S}_w(t, \eta_i) = \left(\mathbf{w}^*\right)^H \hat{\mathbf{S}}(t, \eta_i). \quad (42)$$

The filtered equivalent single-channel echo $\mathbf{S}_w(t, \eta_i)$ is the weighted sum of the echoes transmitted and received by each element. After traversing all the FPIs, the filtered equivalent single-channel echo matrix is acquired. Finally, the jamming-free SAR images can be obtained using the conventional imaging processing algorithm.

Fig. 7 summarizes the complete processing procedure. The contribution of the technique proposed in this article mainly lies in the deception jamming suppression processing of the multichannel data after range compression. A red dashed box highlights the processing steps in the figure.

IV. Simulation Analysis

As an illustration of the validity of the proposed technique, this section provides simulation results. A ground area with multiple targets as the simulation scene is shown in Fig. 8. Set the coordinate origin as the scene center. The size of the imaging area is 620 m × 460 m. SAR works in side-looking stripmap mode. The jammer is template-based and inserts false vehicle targets in the scene via convolutional modulation [5]. The deception jamming template is given as Fig. 9. Table I gives the parameters applied in our simulation.

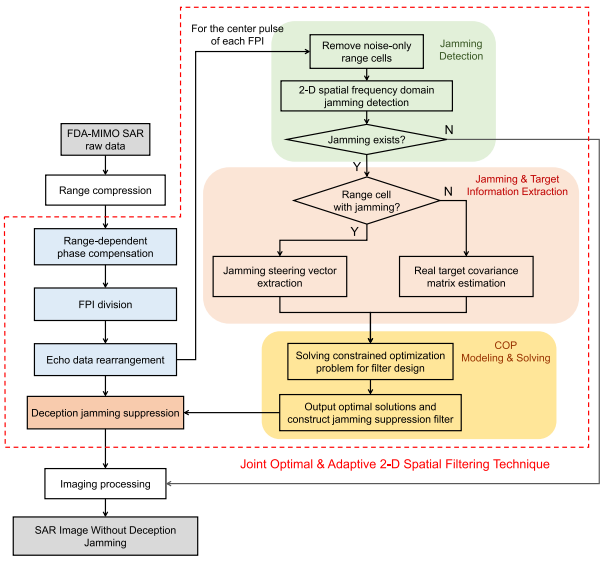


Fig. 7. Flowchart of FDA-MIMO SAR deception jamming suppression processing.

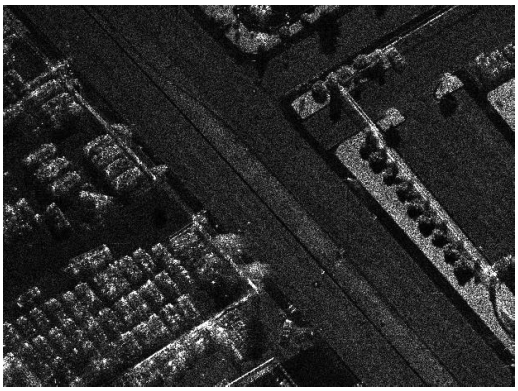


Fig. 8. Original simulation scene.

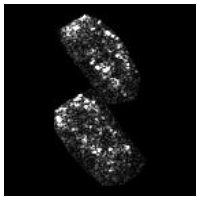


Fig. 9. Deception jamming template.

A. Deception Jamming Detection Simulation Results

This section presents the results of the 2-D spatial frequency domain spectrum estimation before and after range-dependent phase compensation as well as the deception jamming detection result of the simulation scene echo.

Fig. 10 shows the estimation result of the 2-D transmit–receive spatial power spectrum of the echo at the center moment of an FPI. The width of the sliding window is set to 35 range cells, and the step size is 1. Fig. 10(a) depicts the 2-D spatial–spectral distribution of the echo before phase compensation. The magnified image of the 2-D spatial power spectrum of the echo after range-dependent phase compensation is given as Fig. 10(b). Since the range compression peaks of the real targets can be accurately compensated, the real scene echo energy of different range cells is accumulated on

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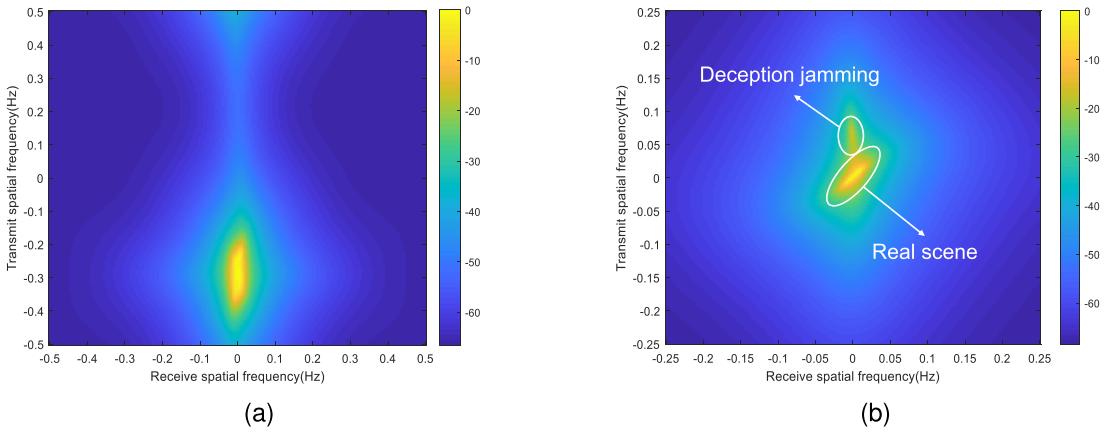


Fig. 10. 2-D spatial power spectrum estimation results. (a) Before phase compensation. (b) Magnified image after phase compensation.

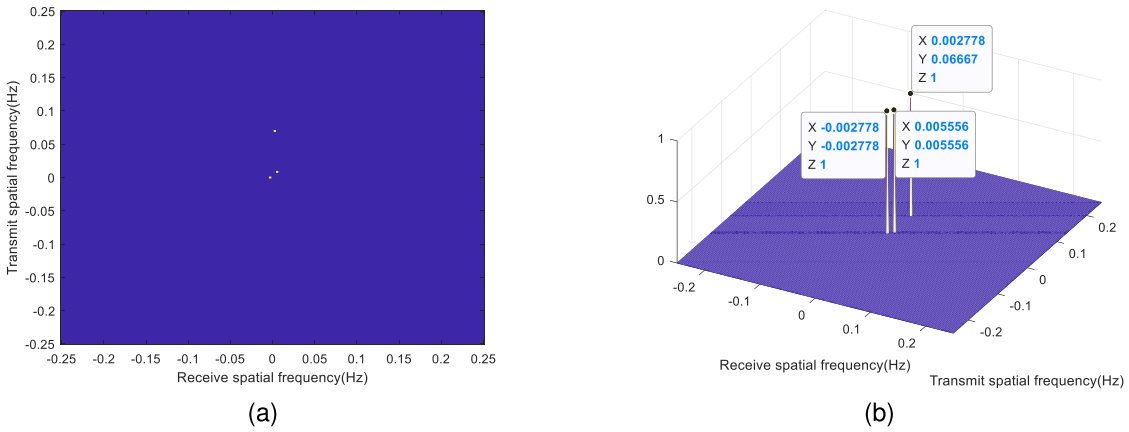


Fig. 11. Jamming detection results based on morphological grayscale reconstruction. (a) Top view. (b) 3-D map.

TABLE I
SIMULATION PARAMETERS

Parameters	Symbols	Values
Pulse Repetition Frequency	f_{PRF}	1000Hz
Frequency Increment	Δf	40kHz
Reference Carrier Frequency	f_0	10GHz
Number of Transmit Elements	N	4
Number of Receive Elements	M	4
Velocity of the Platform	v	200m/s
Element Spacing	d	0.015m
Signal Bandwidth	B	$150MHz$
Signal Pulse Duration	T_p	$1\mu s$
Synthetic Aperture Time	T_s	$1s$
Azimuth Beam Width	θ_{bw}	3°
Platform Position	(x_r, y_r, z_r)	$(7000, 0, 5000)\,m$
False Target Center Position	(x_f, y_f, z_f)	$(-35, 130, 0)\,m$
Jammer Position	(x_j, y_j, z_j)	$(280, -220, 0)\,m$

the diagonal. However, the false targets are area targets modulated by the template, whose echoes span several range cells. Therefore, the spectrum of the compensated false targets has a slight extension in the transmit spatial frequency dimension.

Morphological grayscale reconstruction is performed on the 2-D spectrogram shown in Fig. 10(b). Fig. 11 illustrates the output binarization diagram of the jamming detection processing. The results show that even if the spectral peak energy of jamming is much weaker than the real scene, it is detected as the regional maxima without setting any threshold manually. In Fig. 11, the three regional maxima correspond to the peaks of the real target scene and the jamming targets. According to Fig. 11(b), the 2-D coordinates of the peak point of the real scene are equal, which conforms to the theoretical analysis. By further judging the coordinates of the detected peak points, the deception jamming position can be screened out to calculate the steering vector.

B. Filter Design Simulation Results

After the deception jamming detection of all the range cells, the obtained jamming steering vector and the covariance matrix of the real target scene are substituted into the COP of (39). The constant error tolerance ξ is set to be 10^{-6} and the upper limit ν is set to be 10^{-3}. For the DE algorithm, N_P is set to 100, F_s is set to 0.6, C_r is set to 0.9, and the maximum iteration time G_m is set to 100. The boundaries of the variables to be optimized are set as $[-5, 5]$. Fig. 12 shows the 2-D transmit–receive spatial frequency filter response obtained by solving the COP. The filter forms a deep notch at the deception jamming position, and the output gain of the real target area is guaranteed.

To further demonstrate the filter performance, the jamming notch in the filter response is sliced along the axis, and the profiles are shown in Fig. 13. According to the longitudinal section along the transmit spatial frequency dimension in

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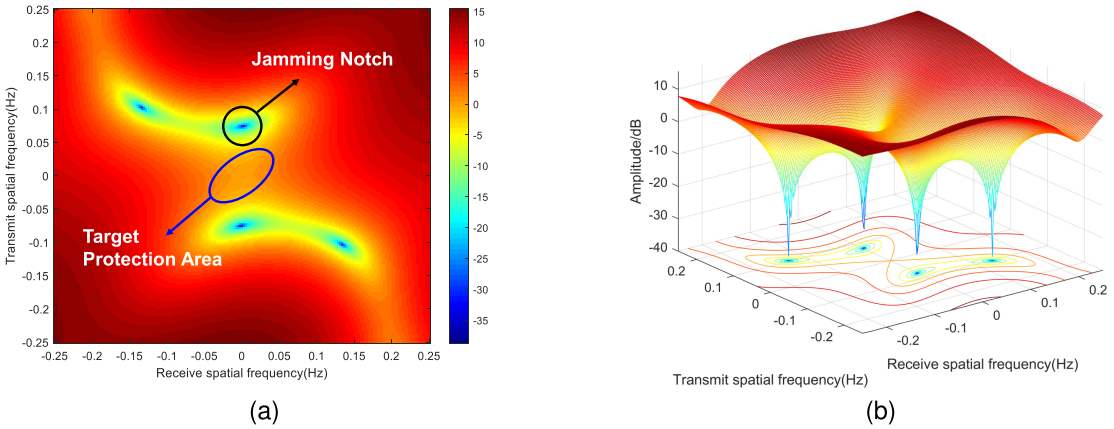


Fig. 12. Filter response of the proposed method. (a) Top view. (b) 3-D map.

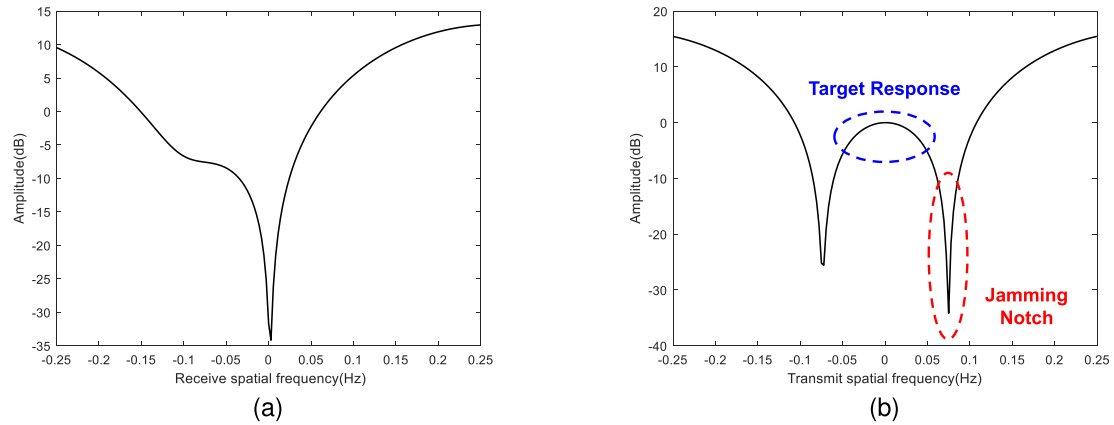


Fig. 13. Longitudinal sections of the designed filter response. (a) Longitudinal section along the receive spatial frequency dimension. (b) Longitudinal section along the transmit spatial frequency dimension.

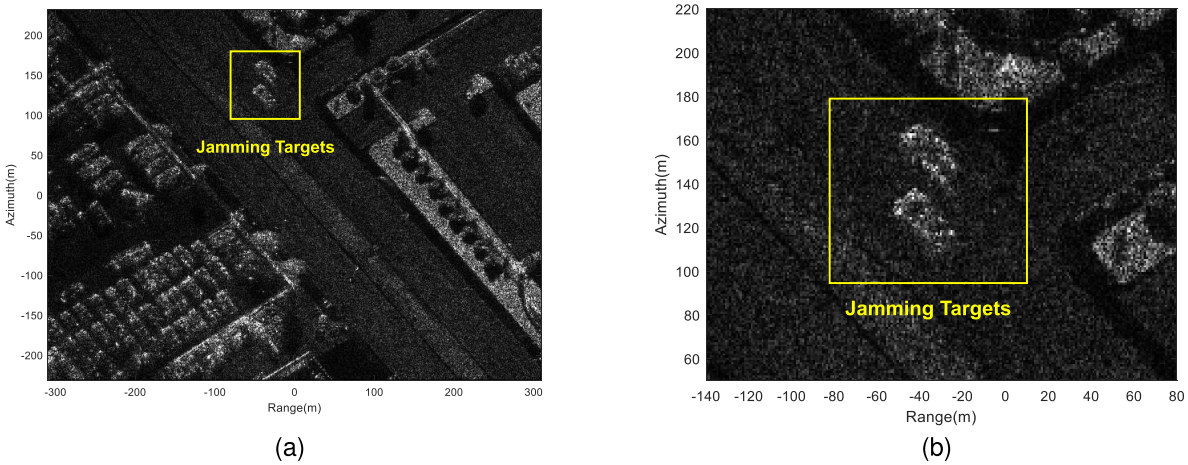


Fig. 14. SAR imaging result of the reference channel disturbed by deception jamming. (a) Full image. (b) Magnified image of the jammed area.

Fig. 13(b), the filter ensures an output gain of about 0 dB at the real target scene area, while a notch of −34.17 dB is formed at the detected jamming position.

C. Deception Jamming Suppression Simulation Results

This section presents the simulation results of SAR imaging based on the proposed deception jamming suppression method.

In this article, we use the back projection (BP) algorithm [39] for SAR imaging processing. The deceptively jammed SAR image of the reference channel is given as Fig. 14. Due to the addition of the deception jamming, the imaging result has two more false vehicle targets than the original scene. Taking 100 PRIs as an FPI, all the echoes are segmented and filtered by the proposed 2-D spatial filtering technique. The SAR imaging result after jamming suppression processing is shown in Fig. 15. After processing with the proposed method, the jamming targets are effectively suppressed. Figs. 14(b) and 15(b) illustrate the magnification of

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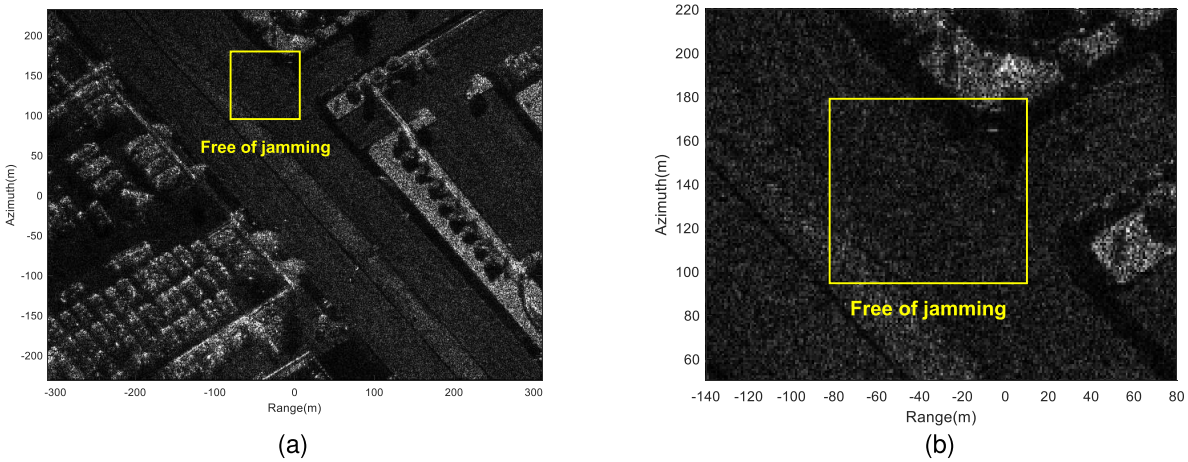


Fig. 15. SAR imaging result after jamming suppression processing. (a) Full image. (b) Magnified image of the jammed area.

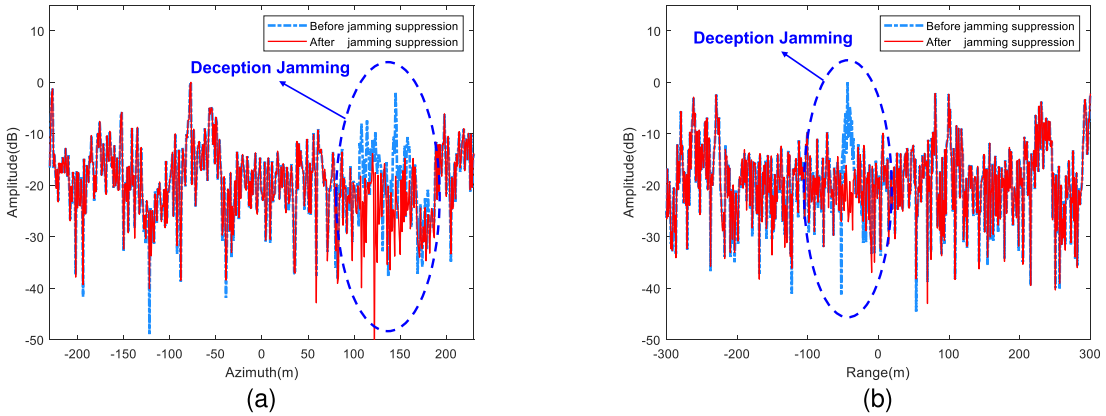


Fig. 16. Longitudinal sections of the SAR image before and after processing. (a) Longitudinal section along the azimuth dimension. (b) Longitudinal section along the range dimension.

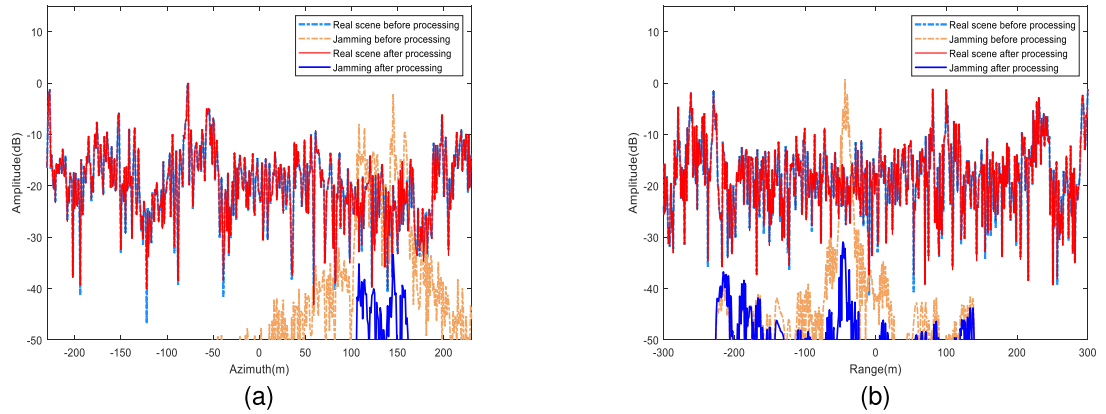


Fig. 17. Comparison of real scene and jamming targets before and after processing. (a) Longitudinal section along the azimuth dimension. (b) Longitudinal section along the range dimension.

the jamming area before and after filtering. The comparison shows that the filtering of the deception jamming does not lose the information of the covered scatterers on the ground.

The images are sliced along the range and azimuth dimensions to further analyze the jamming suppression performance. Fig. 16 presents the before-and-after comparison profiles along $x = -33$ m and $y = 132$ m, respectively. Protected by the output gain constraint, the real targets are almost wholly preserved after processing, while the deception jamming is significantly weakened. Fig. 17 shows the amplitude comparison of the real scene and jamming targets before and after filtering to verify the respective filtering effects of deception jamming and real scene echo. Since the detected jamming steering vector is the peak of the power spectrum, the filter solved by this method has a considerable suppression effect in the peak area of the jamming targets. The curves of the real scene before and after filtering almost completely overlap.

To give a numerical quantification of the performance, the ratio of the average power of the real target scene to the average power of the false targets in the jamming area

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of the SAR image is defined as the signal-to-jamming ratio (SJR), which is given by

$$SJR = 10\log_{10}\frac{E[P_t]}{E\left[P_f\right]} \tag{43}$$

where P_f is the power of the false target image pixel, and P_t is the power of the real scene image pixel in the jammed area. The peak power of the deception jamming is reduced by 31.7 dB based on the processing procedure recommended in this article, while the peak power of the real scene is reduced by 0.01 dB. The SJR of the output SAR image is increased from −18.94 to 5.15 dB, i.e., the improvement of 24.09 dB is realized.

V. Conclusion

This article proposes an approach to suppress deception jamming by utilizing the characteristics of FDA-MIMO SAR. Based on the properties of deception jamming and real target scene in the time and 2-D transmit–receive spatial frequency domains, the range-dependent phase is compensated to construct separability for filtering. The peak position of the deception jamming and the covariance matrix of the real scene are adaptively extracted based on the jamming detection and localization strategy without prior knowledge. The 2-D spatial filter design task is modeled as a COP to get the desired filter shape. The proposed technique exhibits satisfactory performance when the preset position of false targets differs from the jammer, and the forwarding delay of the jammer is not required to exceed one PRI. It is verified by the simulation experiment scene that there is a 24.09-dB improvement in image SJR compared to the image before jamming suppression.

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